Pikachu Metrics 数据分析 metrics_Mjlab-Velocity-Flat-Pikachu-Quad__model_4999_20260826_170813_with_target

FILE
metrics_Mjlab-Velocity-Flat-Pikachu-Quad__model_4999_20260826_170813_with_target.csv

NETWORK
Mjlab-Velocity-Flat-Pikachu-Quad__model_4999

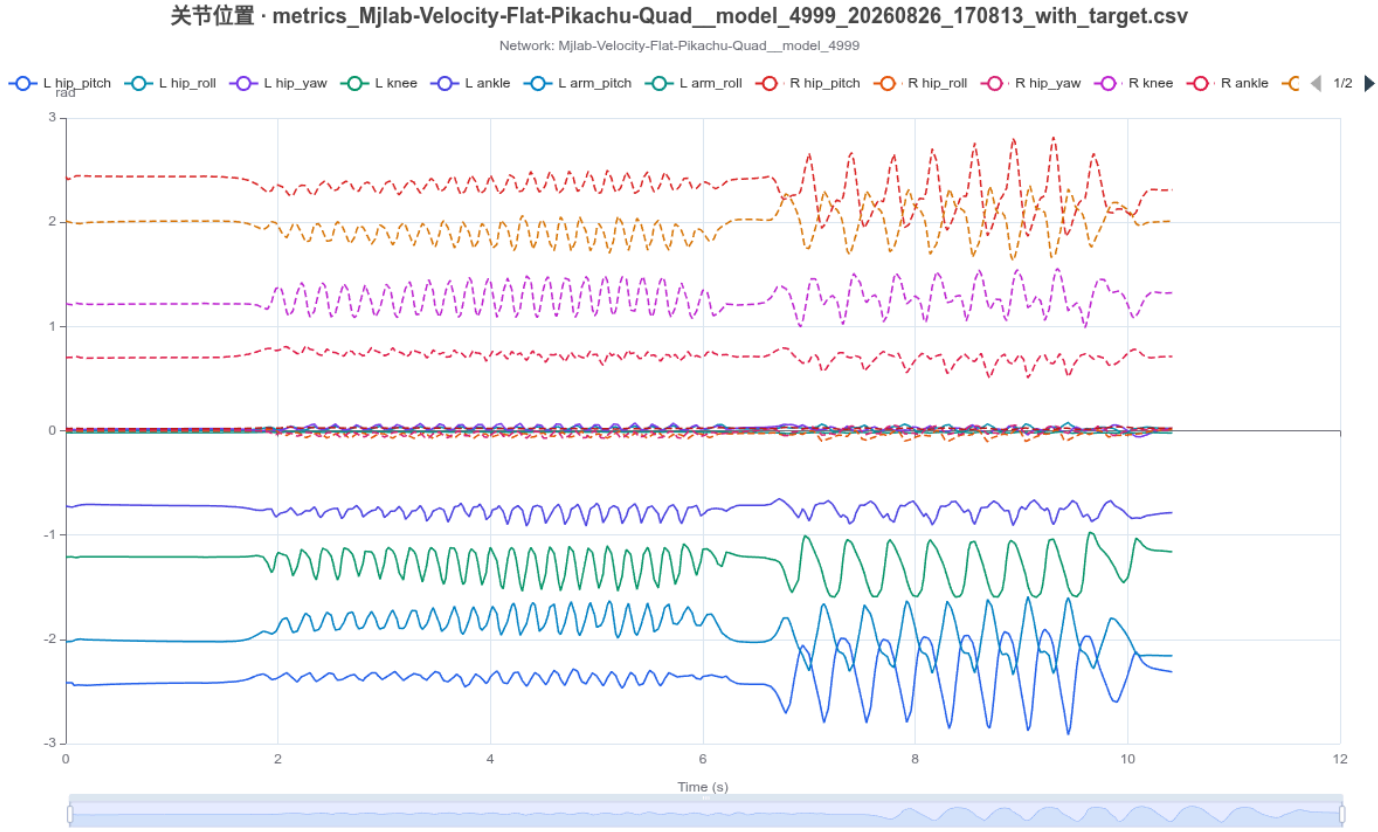
SAMPLES
1,031

DURATION
22.760 s

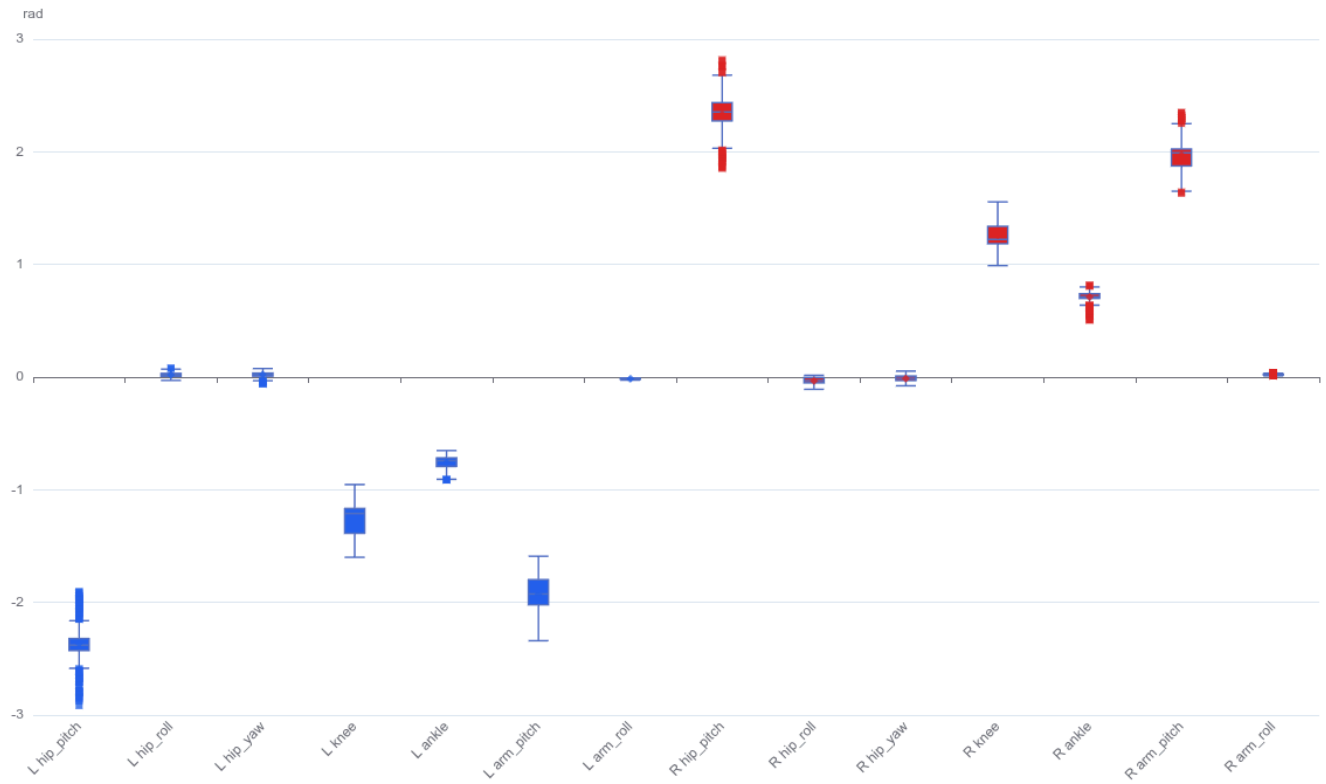
ENVIRONMENTS
1

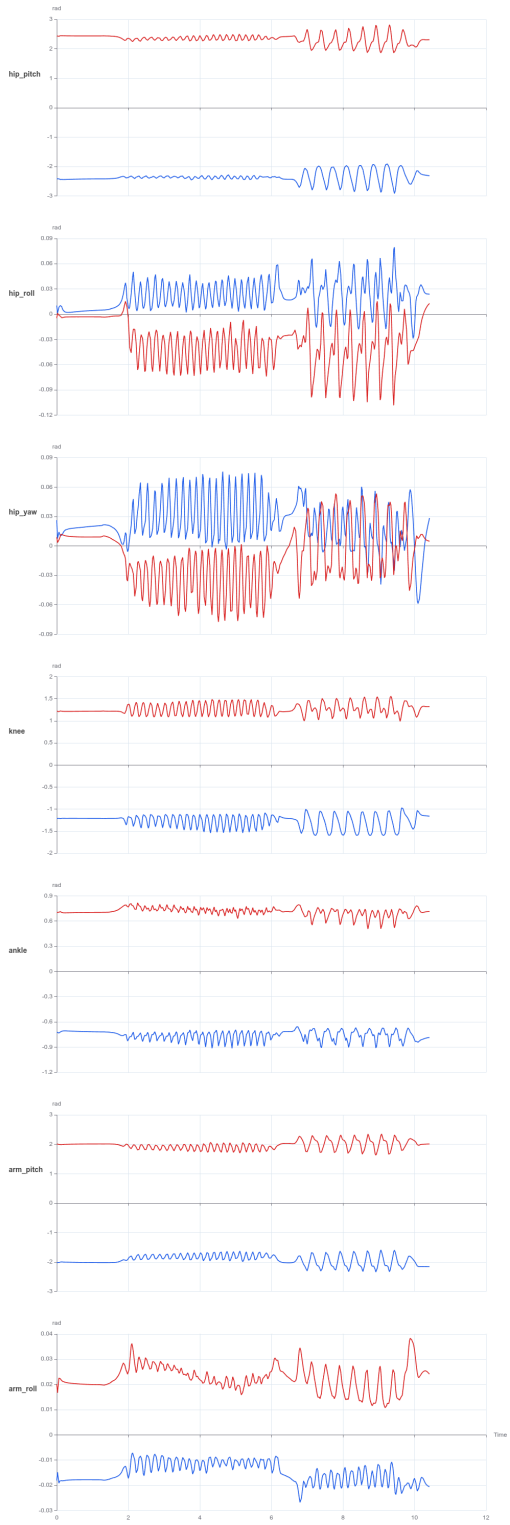
时间区间: 0.000 – 10.420 s · 仅含已勾选字段

关节位置

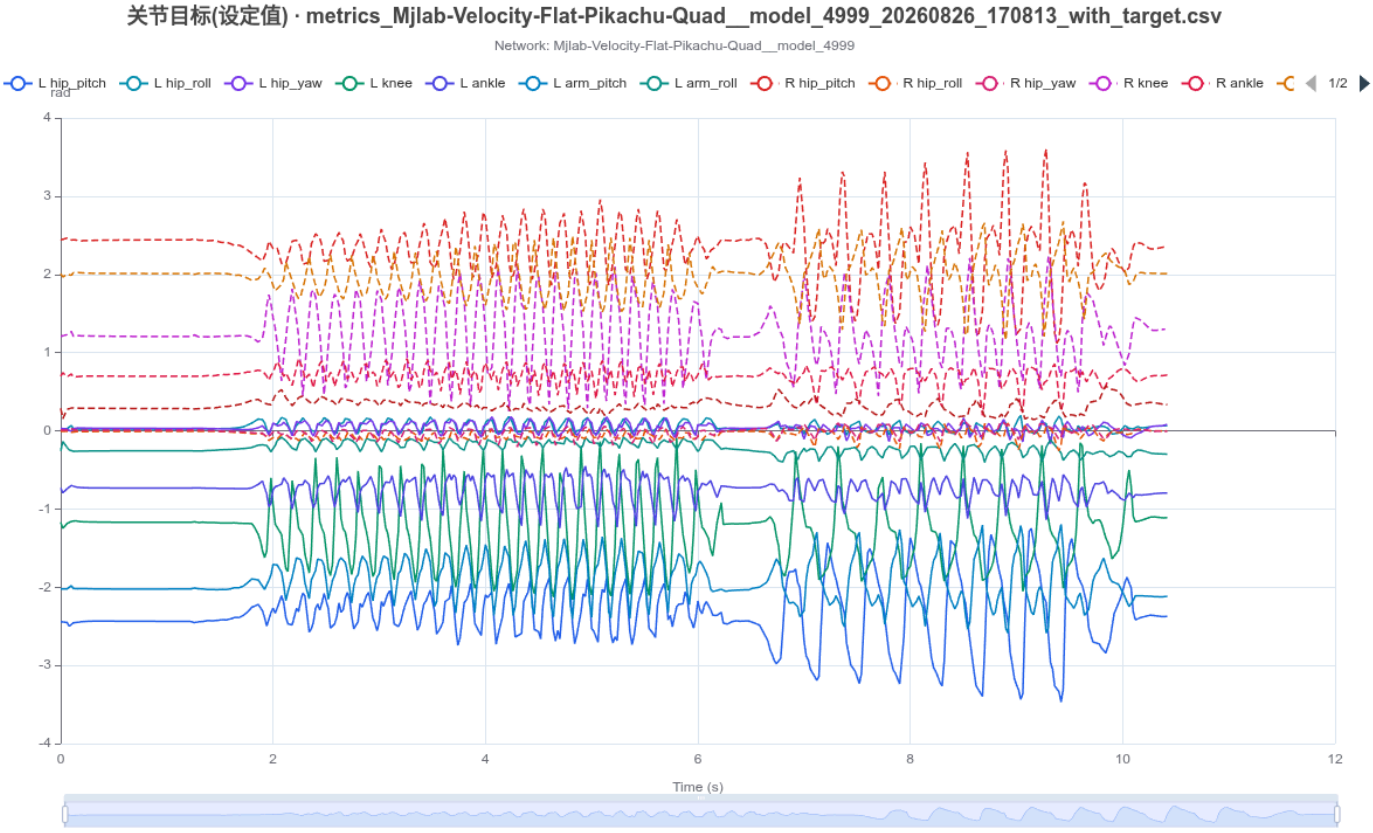


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	470	-2.91542	-1.90278	-2.36309	2.36950	2.91542	2.67435
joint_position_rad/left_hip_roll_joint	10	15	470	-0.0288299	0.0795395	0.0209791	0.0271006	0.0795395	0.0472199
joint_position_rad/left_hip_yaw_joint	10	15	470	-0.0588003	0.0756450	0.0226612	0.0320302	0.0756450	0.0631507
joint_position_rad/left_knee_joint	30	10	470	-1.59978	-0.953836	-1.27270	1.28171	1.59978	1.57655
joint_position_rad/left_ankle_joint	10	15	470	-0.913993	-0.651727	-0.757501	0.759549	0.913993	0.875356
joint_position_rad/left_arm_pitch_joint	10	15	470	-2.34033	-1.59042	-1.92903	1.93591	2.34033	2.17721
joint_position_rad/left_arm_roll_joint	10	15	470	-0.0267989	-0.00718159	-0.0150834	0.0156225	0.0267989	0.0211315
joint_position_rad/right_hip_pitch_joint	30	10	470	1.85322	2.81676	2.33219	2.33743	2.81676	2.52638
joint_position_rad/right_hip_roll_joint	10	15	470	-0.108398	0.0154863	-0.0327998	0.0420109	0.108398	0.0743195
joint_position_rad/right_hip_yaw_joint	10	15	470	-0.0775052	0.0532860	-0.0122536	0.0312478	0.0775052	0.0634576
joint_position_rad/right_knee_joint	30	10	470	0.988865	1.55503	1.26031	1.26589	1.55503	1.47202
joint_position_rad/right_ankle_joint	10	15	470	0.505881	0.815512	0.710960	0.712687	0.815512	0.779166
joint_position_rad/right_arm_pitch_joint	10	15	470	1.63169	2.35019	1.96972	1.97466	2.35019	2.21119
joint_position_rad/right_arm_roll_joint	10	15	470	0.0107571	0.0382577	0.0224034	0.0229445	0.0382577	0.0301086

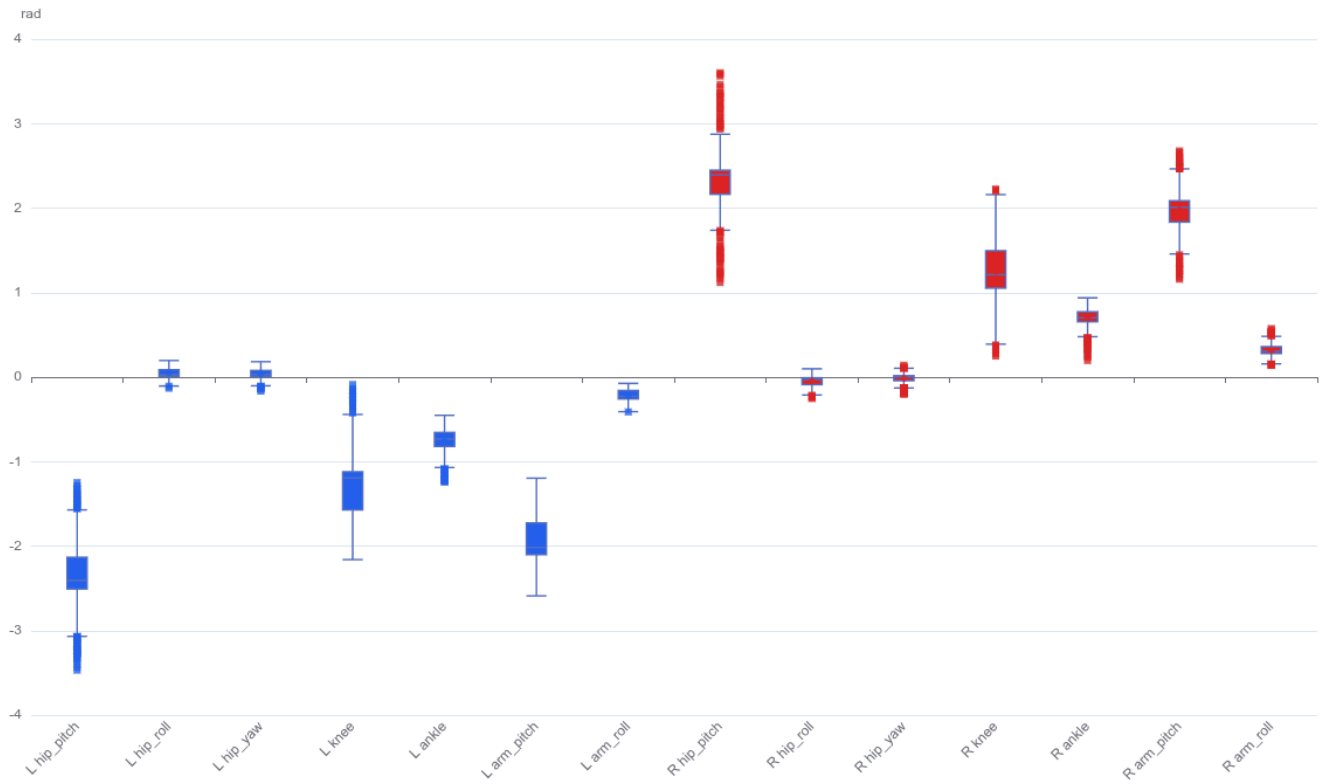

rad
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

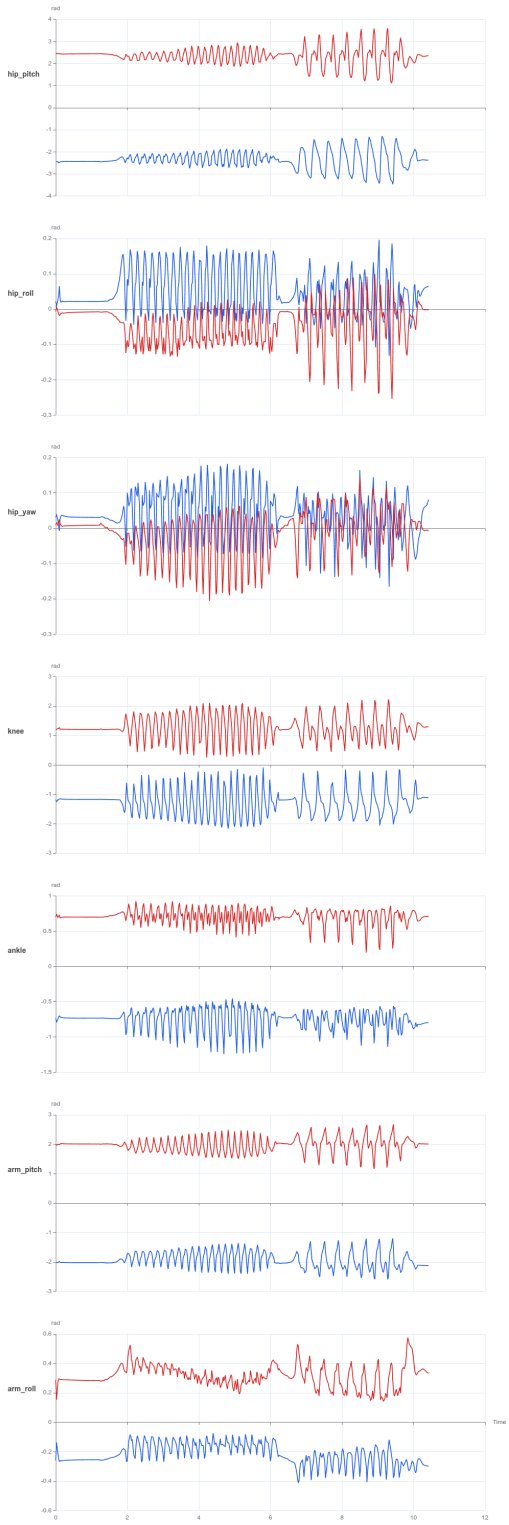

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节目标(设定值)

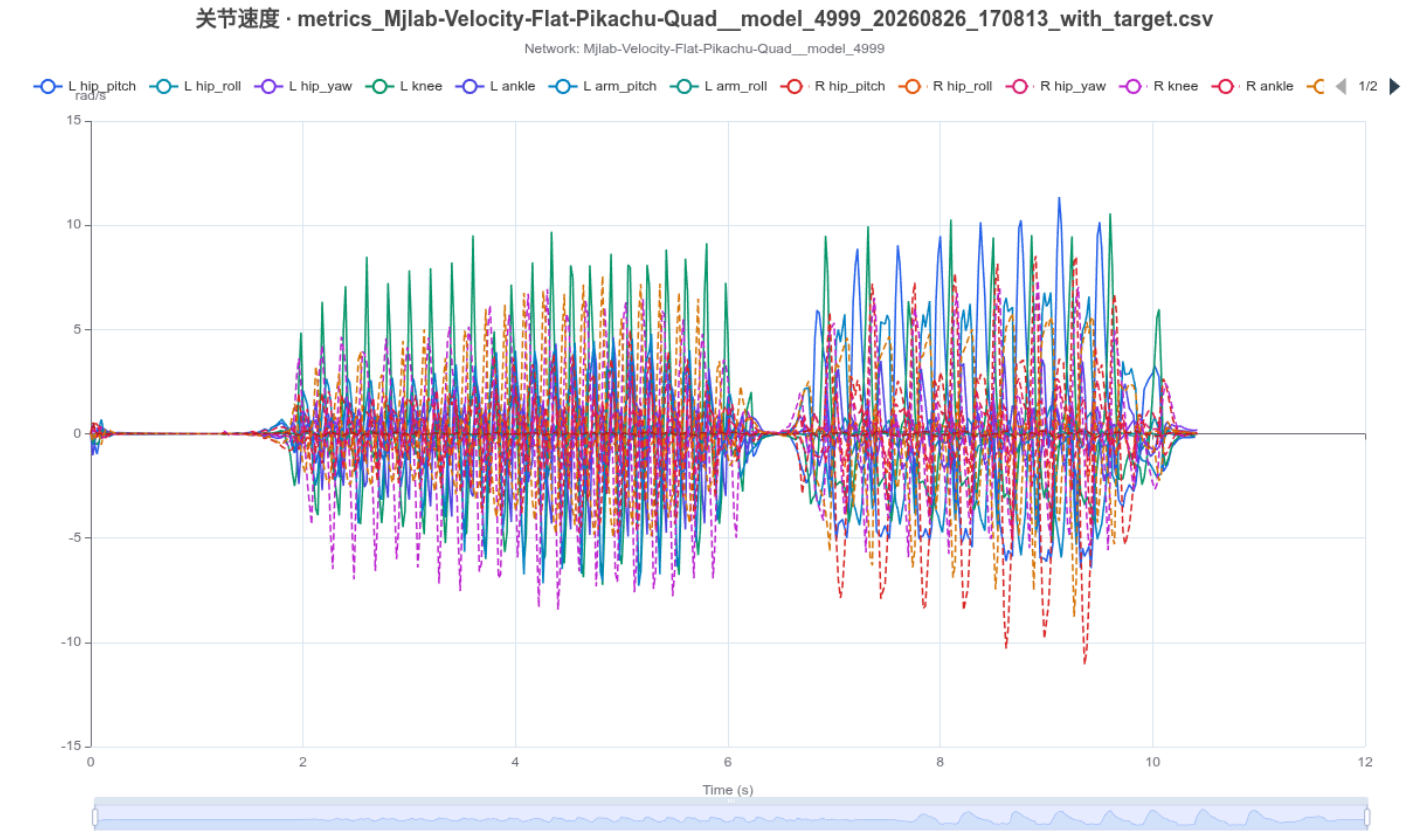


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	470	-3.46823	-1.24874	-2.35249	2.38459	3.46823	3.10193
joint_target_rad/left_hip_roll_joint	10	15	470	-0.132076	0.196193	0.0485176	0.0765114	0.196193	0.159235
joint_target_rad/left_hip_yaw_joint	10	15	470	-0.165667	0.182138	0.0379642	0.0718800	0.182138	0.141225
joint_target_rad/left_knee_joint	30	10	470	-2.15983	-0.0878150	-1.26460	1.33929	2.15983	1.94677
joint_target_rad/left_ankle_joint	10	15	470	-1.24366	-0.452916	-0.752709	0.766274	1.24366	1.02742
joint_target_rad/left_arm_pitch_joint	10	15	470	-2.58862	-1.19624	-1.93260	1.95098	2.58862	2.35200
joint_target_rad/left_arm_roll_joint	10	15	470	-0.412045	-0.0744632	-0.219495	0.230818	0.412045	0.348291
joint_target_rad/right_hip_pitch_joint	30	10	470	1.11978	3.60145	2.33668	2.36530	3.60145	2.93503
joint_target_rad/right_hip_roll_joint	10	15	470	-0.254101	0.0986963	-0.0456384	0.0720720	0.254101	0.127689
joint_target_rad/right_hip_yaw_joint	10	15	470	-0.205907	0.141246	-0.0155148	0.0639144	0.205907	0.141822
joint_target_rad/right_knee_joint	30	10	470	0.248905	2.22566	1.24977	1.31145	2.22566	1.96975
joint_target_rad/right_ankle_joint	10	15	470	0.196218	0.937561	0.701663	0.710193	0.937561	0.858801
joint_target_rad/right_arm_pitch_joint	10	15	470	1.15862	2.67676	1.97841	1.99429	2.67676	2.41424
joint_target_rad/right_arm_roll_joint	10	15	470	0.139089	0.574471	0.313055	0.322110	0.574471	0.428534

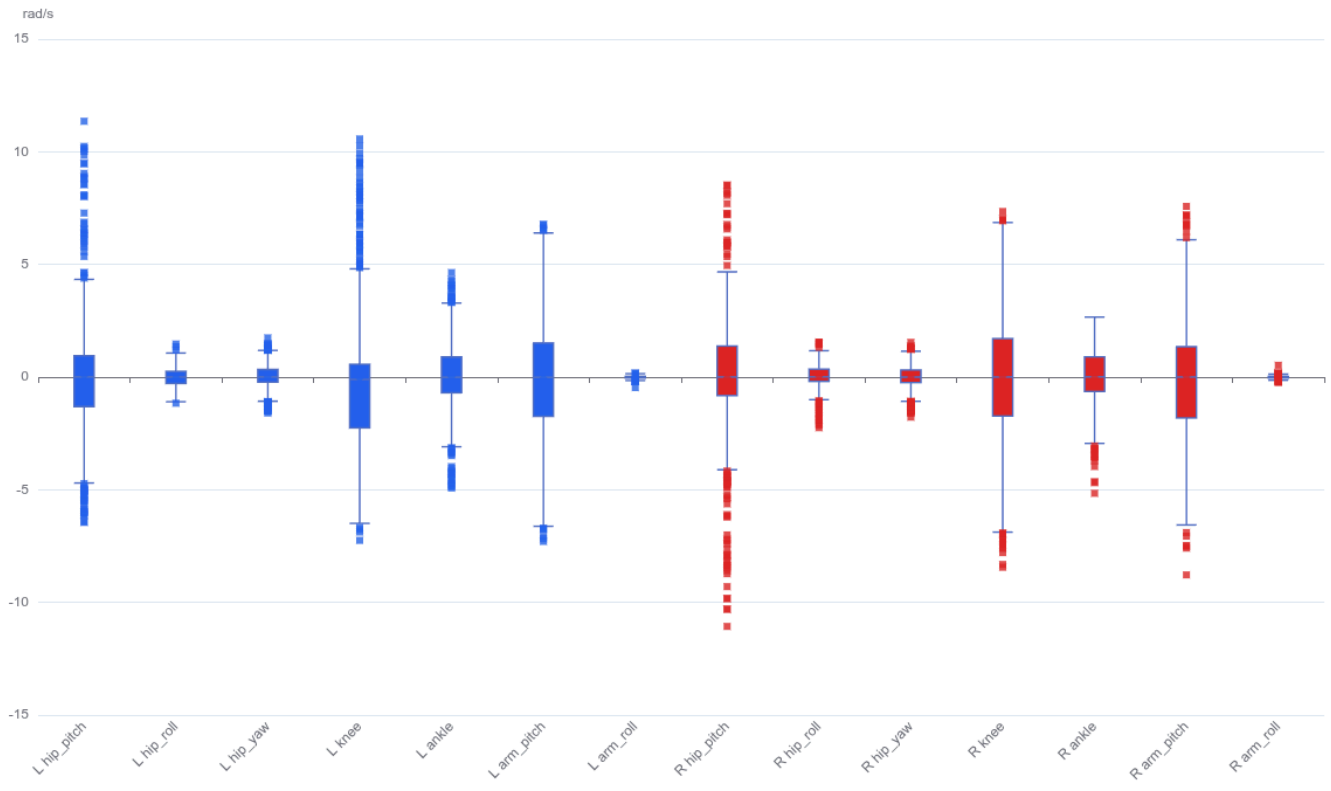

rad
4
3
2
1
0
-1
-2
-3
-4
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

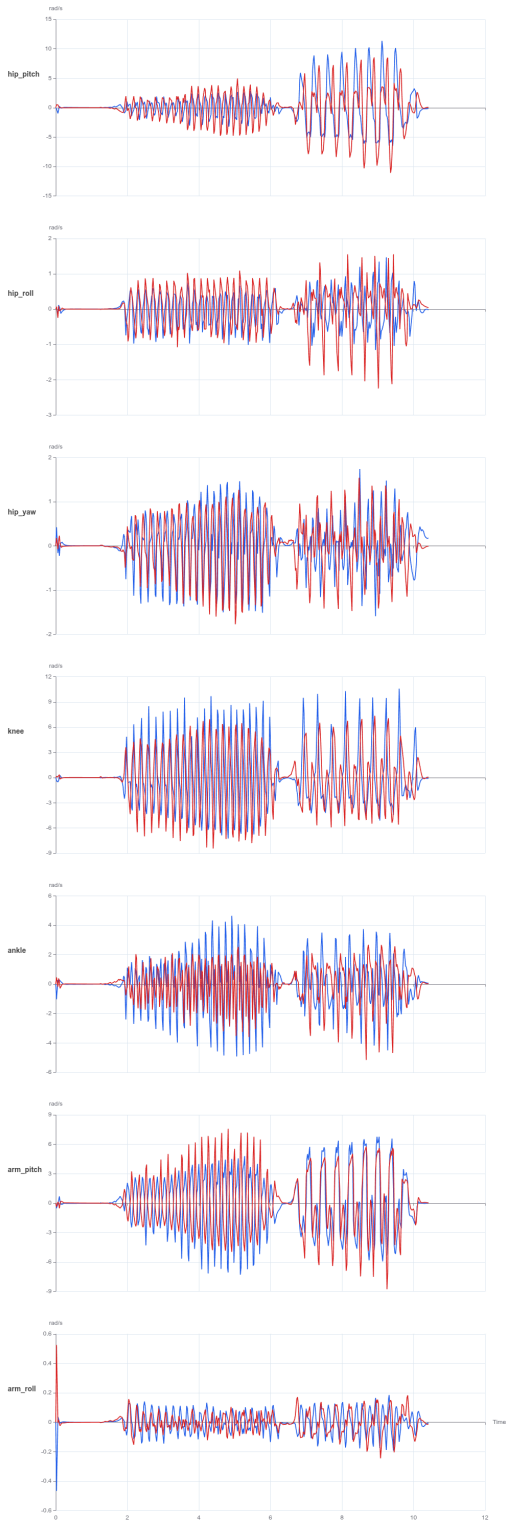

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

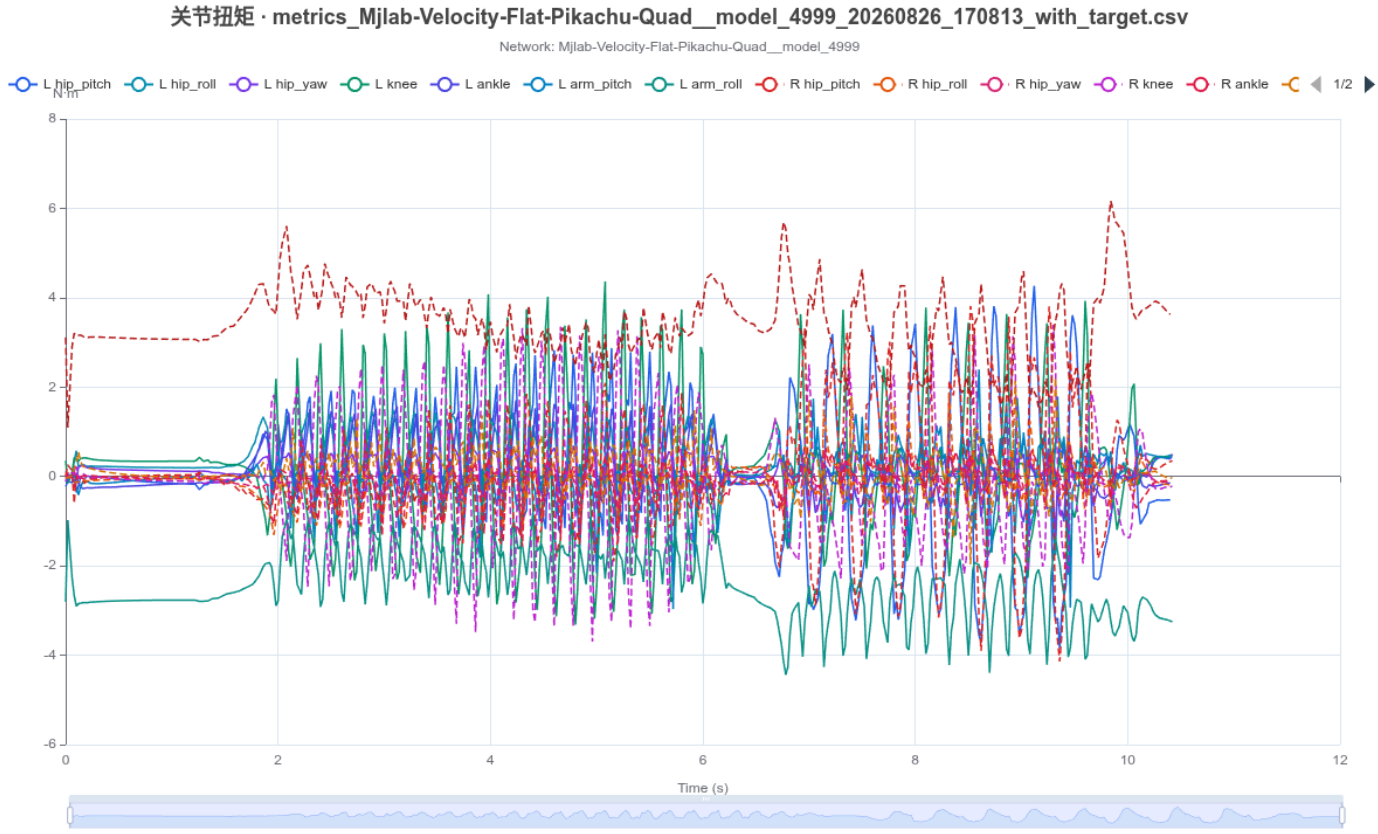


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	470	-6.44869	11.3546	0.0637101	2.95852	11.3546	6.40038
joint_velocity_rad_s/left_hip_roll_joint	10	15	470	-1.15905	1.46815	-0.0141740	0.420675	1.46815	0.823521
joint_velocity_rad_s/left_hip_yaw_joint	10	15	470	-1.59401	1.74021	0.00604939	0.603079	1.74021	1.27777
joint_velocity_rad_s/left_knee_joint	30	10	470	-7.24698	10.5694	-0.0513693	3.45140	10.5694	7.73758
joint_velocity_rad_s/left_ankle_joint	10	15	470	-4.92235	4.63673	0.0117452	1.65209	4.92235	3.49665
joint_velocity_rad_s/left_arm_pitch_joint	10	15	470	-7.29271	6.77305	-0.0595548	2.87128	7.29271	5.95084
joint_velocity_rad_s/left_arm_roll_joint	10	15	470	-0.467158	0.184736	-0.00160872	0.0718577	0.467158	0.132250
joint_velocity_rad_s/right_hip_pitch_joint	30	10	470	-11.0656	8.52825	-0.0165574	3.04651	11.0656	7.27128
joint_velocity_rad_s/right_hip_roll_joint	10	15	470	-2.24639	1.55416	0.0257526	0.559396	2.24639	1.05065
joint_velocity_rad_s/right_hip_yaw_joint	10	15	470	-1.77612	1.54296	-0.00597390	0.587016	1.77612	1.27183
joint_velocity_rad_s/right_knee_joint	30	10	470	-8.43993	7.35101	0.00664121	3.25572	8.43993	6.59629
joint_velocity_rad_s/right_ankle_joint	10	15	470	-5.15777	2.66070	-0.00218187	1.28386	5.15777	2.56441
joint_velocity_rad_s/right_arm_pitch_joint	10	15	470	-8.77871	7.57557	0.0116129	2.86910	8.77871	5.70774
joint_velocity_rad_s/right_arm_roll_joint	10	15	470	-0.245056	0.524520	-0.000724516	0.0684983	0.524520	0.130487

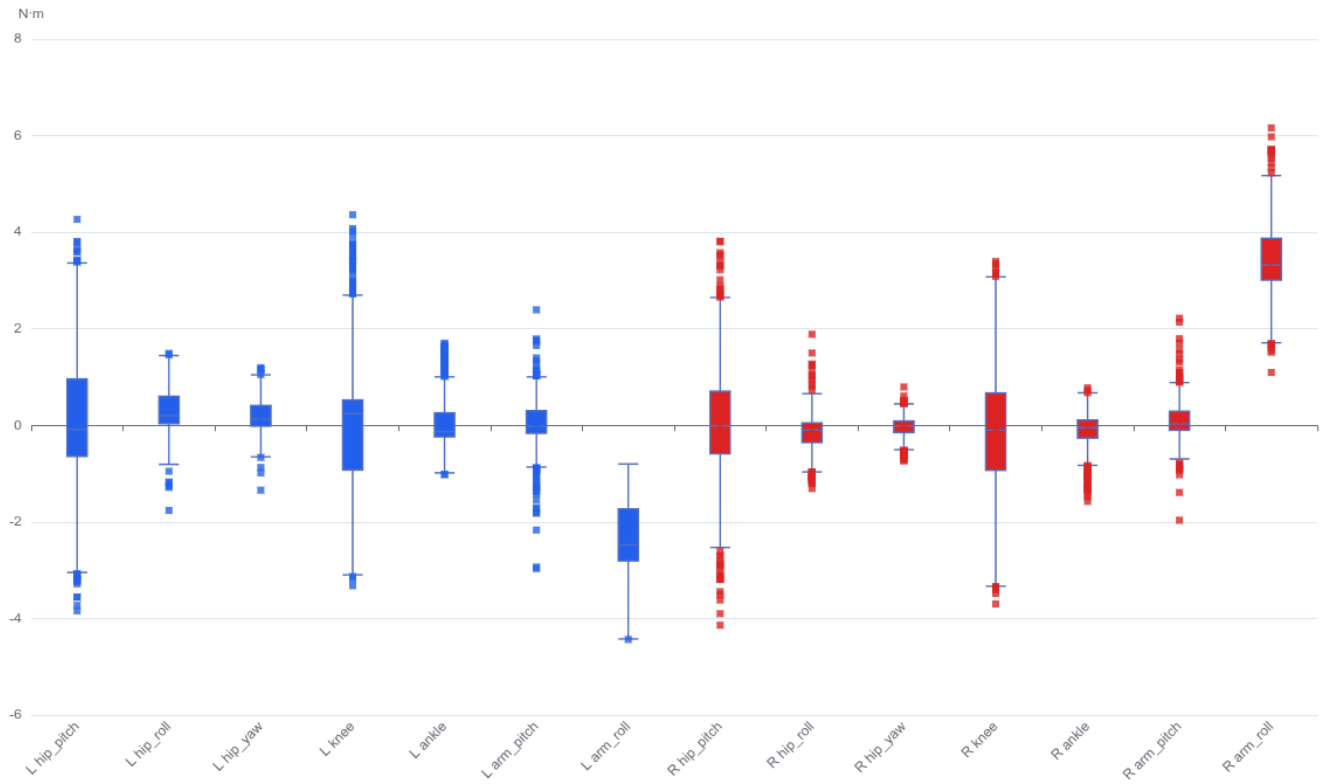

rad/s
15
10
5
0
-5
-10
-15
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

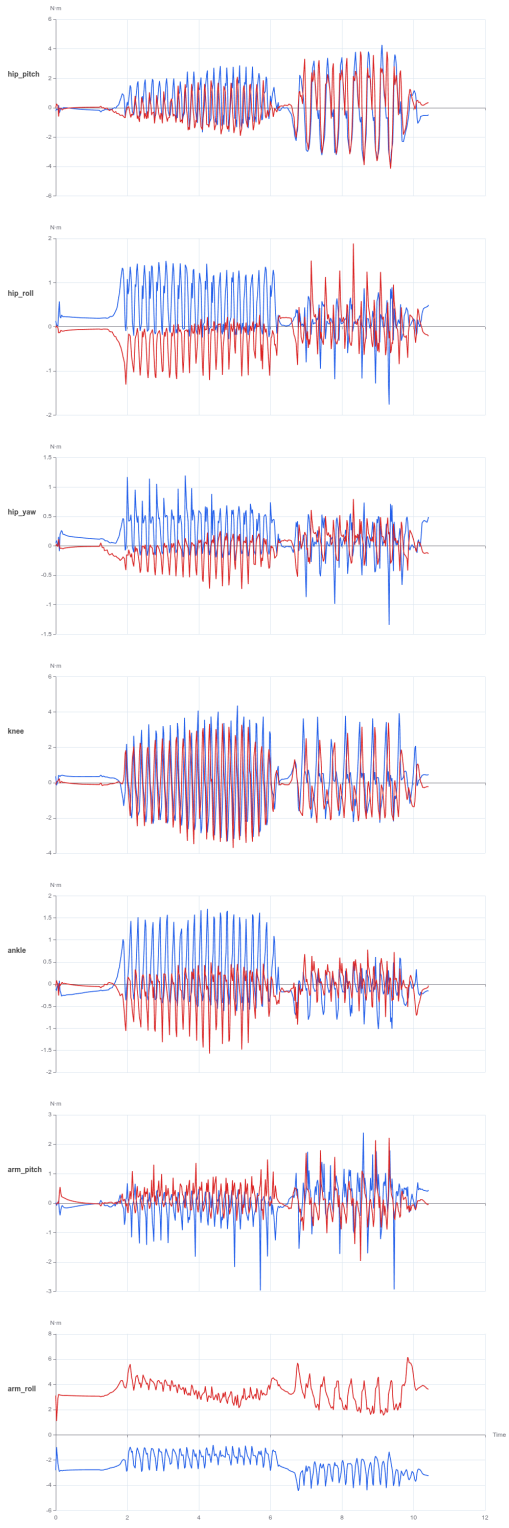

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad/s
Time

关节扭矩



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	470	-3.84153	4.26787	0.0710049	1.47740	4.26787	3.07675
joint_torque_nm/left_hip_roll_joint	10	15	470	-1.76192	1.49165	0.324245	0.578094	1.76192	1.27682
joint_torque_nm/left_hip_yaw_joint	10	15	470	-1.34214	1.19469	0.170467	0.345073	1.34214	0.663312
joint_torque_nm/left_knee_joint	30	10	470	-3.32091	4.36317	0.0814916	1.52278	4.36317	3.29854
joint_torque_nm/left_ankle_joint	10	15	470	-1.01915	1.70405	0.0670256	0.547353	1.70405	1.40145
joint_torque_nm/left_arm_pitch_joint	10	15	470	-2.96609	2.39353	0.00310303	0.558414	2.96609	1.16583
joint_torque_nm/left_arm_roll_joint	10	15	470	-4.43518	-0.800887	-2.38013	2.50083	4.43518	3.72180
joint_torque_nm/right_hip_pitch_joint	30	10	470	-4.13884	3.81383	0.0419660	1.34270	4.13884	2.88757
joint_torque_nm/right_hip_roll_joint	10	15	470	-1.30837	1.88672	-0.151747	0.445672	1.88672	1.04082
joint_torque_nm/right_hip_yaw_joint	10	15	470	-0.732895	0.794996	-0.0310199	0.219314	0.794996	0.480926
joint_torque_nm/right_knee_joint	30	10	470	-3.69740	3.39343	-0.0864424	1.38598	3.69740	2.87746
joint_torque_nm/right_ankle_joint	10	15	470	-1.57465	0.777476	-0.105916	0.392408	1.57465	0.860833
joint_torque_nm/right_arm_pitch_joint	10	15	470	-1.96499	2.21913	0.0971383	0.441486	2.21913	0.886120
joint_torque_nm/right_arm_roll_joint	10	15	470	1.09621	6.16014	3.38558	3.48195	6.16014	4.59687

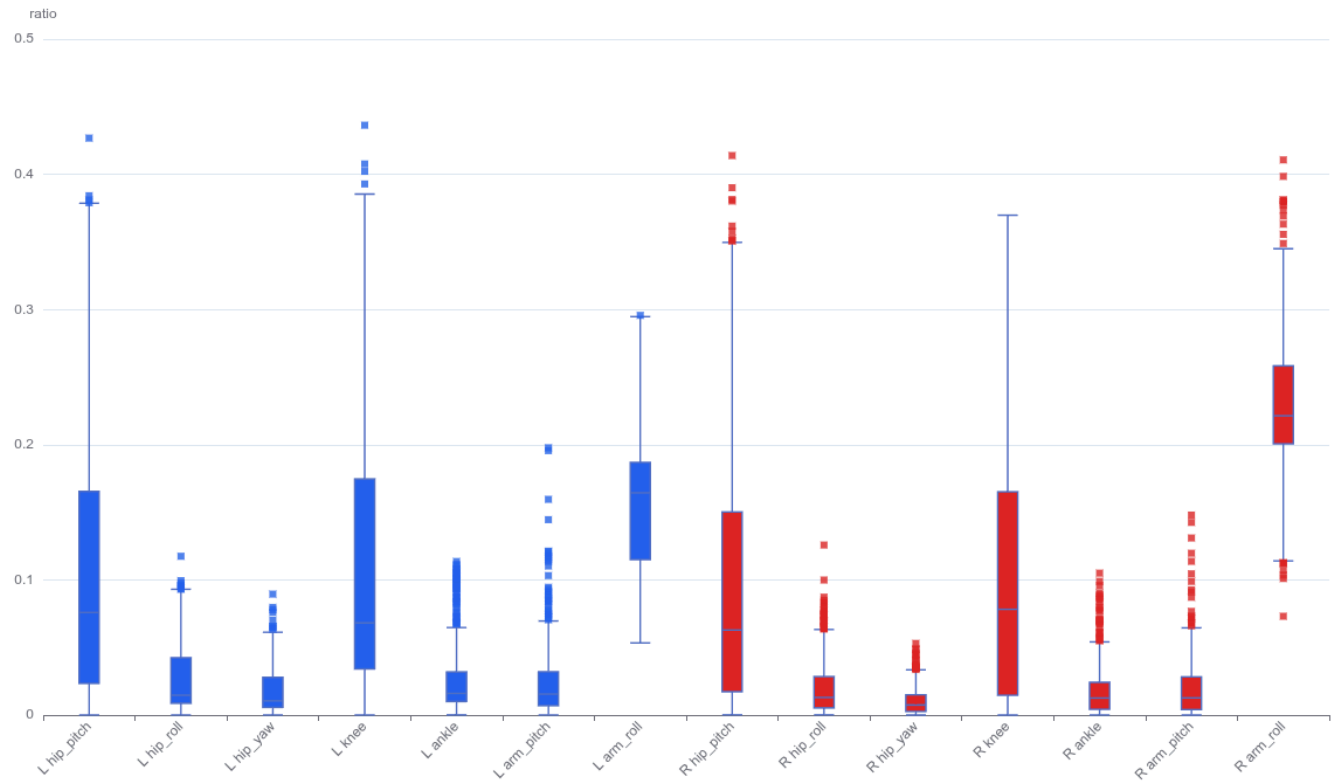

N·m
8
6
4
2
0
-2
-4
-6
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

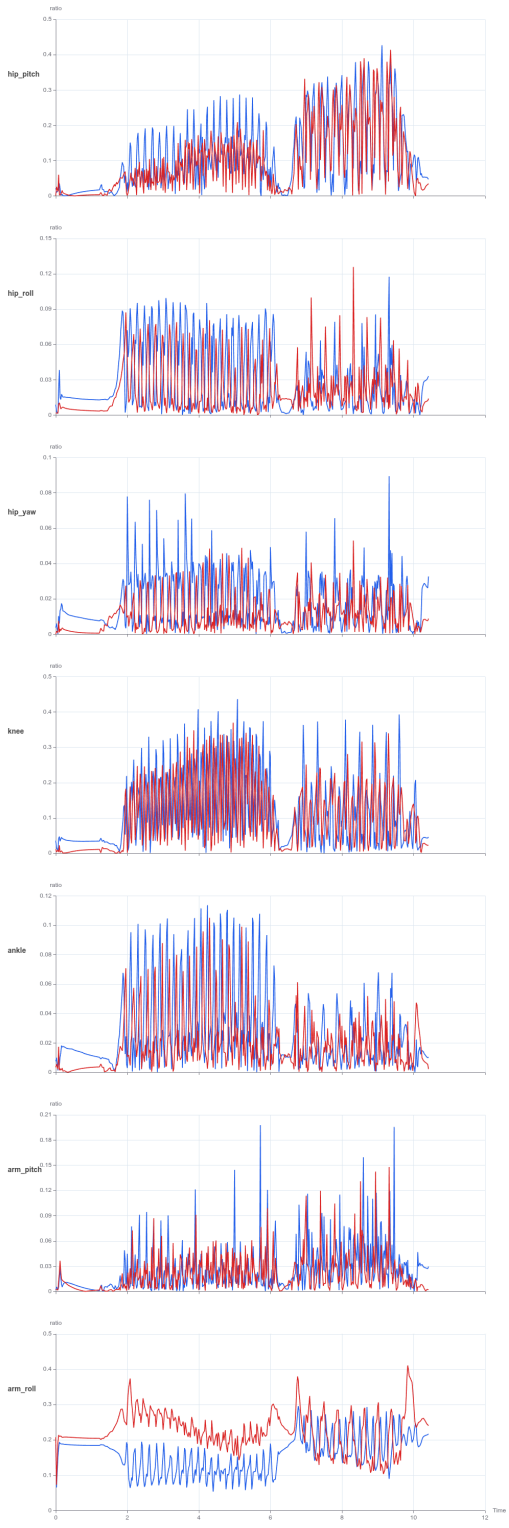

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

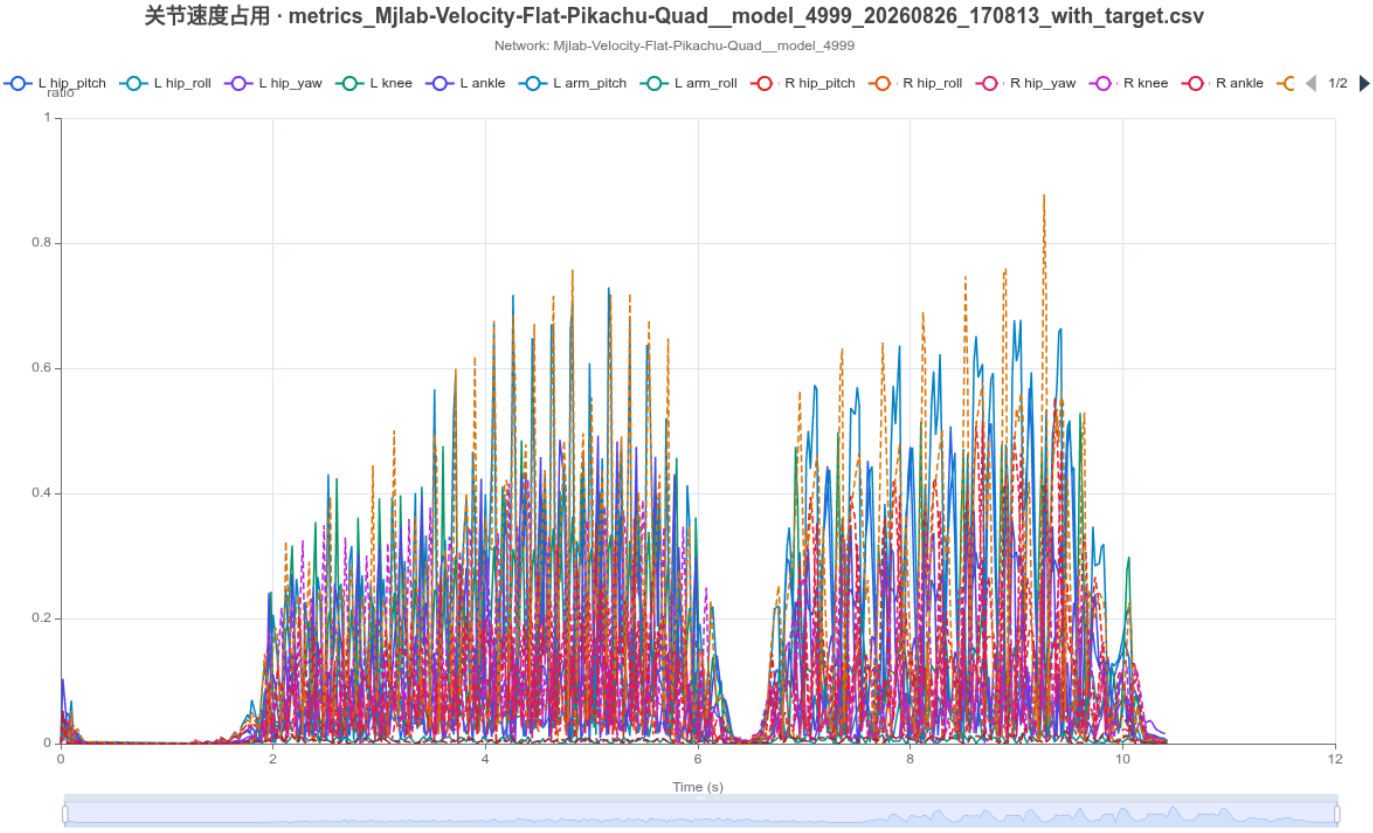


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	470	0.0000203162	0.426787	0.109019	0.147740	0.426787	0.307675
joint_torque_ratio/left_hip_roll_joint	10	15	470	0.0000849287	0.117461	0.0276718	0.0385396	0.117461	0.0851215
joint_torque_ratio/left_hip_yaw_joint	10	15	470	0.0000935263	0.0894761	0.0171121	0.0230049	0.0894761	0.0442208
joint_torque_ratio/left_knee_joint	30	10	470	0.0000443800	0.436317	0.112563	0.152278	0.436317	0.329854
joint_torque_ratio/left_ankle_joint	10	15	470	0.000102229	0.113603	0.0259075	0.0364902	0.113603	0.0934302
joint_torque_ratio/left_arm_pitch_joint	10	15	470	0.0000812686	0.197739	0.0248578	0.0372276	0.197739	0.0777222
joint_torque_ratio/left_arm_roll_joint	10	15	470	0.0533925	0.295679	0.158675	0.166722	0.295679	0.248120
joint_torque_ratio/right_hip_pitch_joint	30	10	470	0.000200901	0.413884	0.0957850	0.134270	0.413884	0.288757
joint_torque_ratio/right_hip_roll_joint	10	15	470	0.000153457	0.125781	0.0209458	0.0297115	0.125781	0.0693877
joint_torque_ratio/right_hip_yaw_joint	10	15	470	0.00000819642	0.0529997	0.0105911	0.0146209	0.0529997	0.0320617
joint_torque_ratio/right_knee_joint	30	10	470	0.00000413733	0.369740	0.101572	0.138598	0.369740	0.287746
joint_torque_ratio/right_ankle_joint	10	15	470	0.0000200863	0.104977	0.0180389	0.0261605	0.104977	0.0573889
joint_torque_ratio/right_arm_pitch_joint	10	15	470	0.00000596826	0.147942	0.0197238	0.0294324	0.147942	0.0590747
joint_torque_ratio/right_arm_roll_joint	10	15	470	0.0730805	0.410676	0.225705	0.232130	0.410676	0.306458

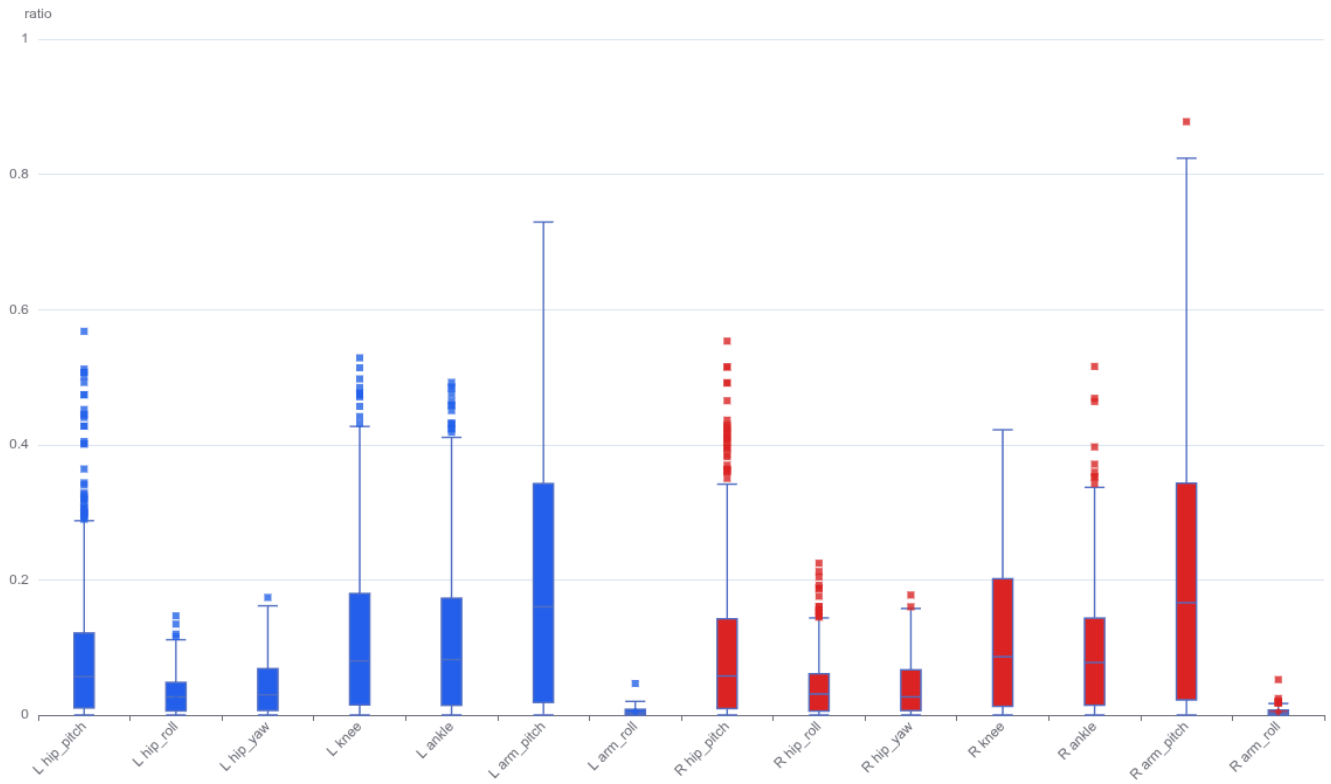

ratio
0.5
0.4
0.3
0.2
0.1
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

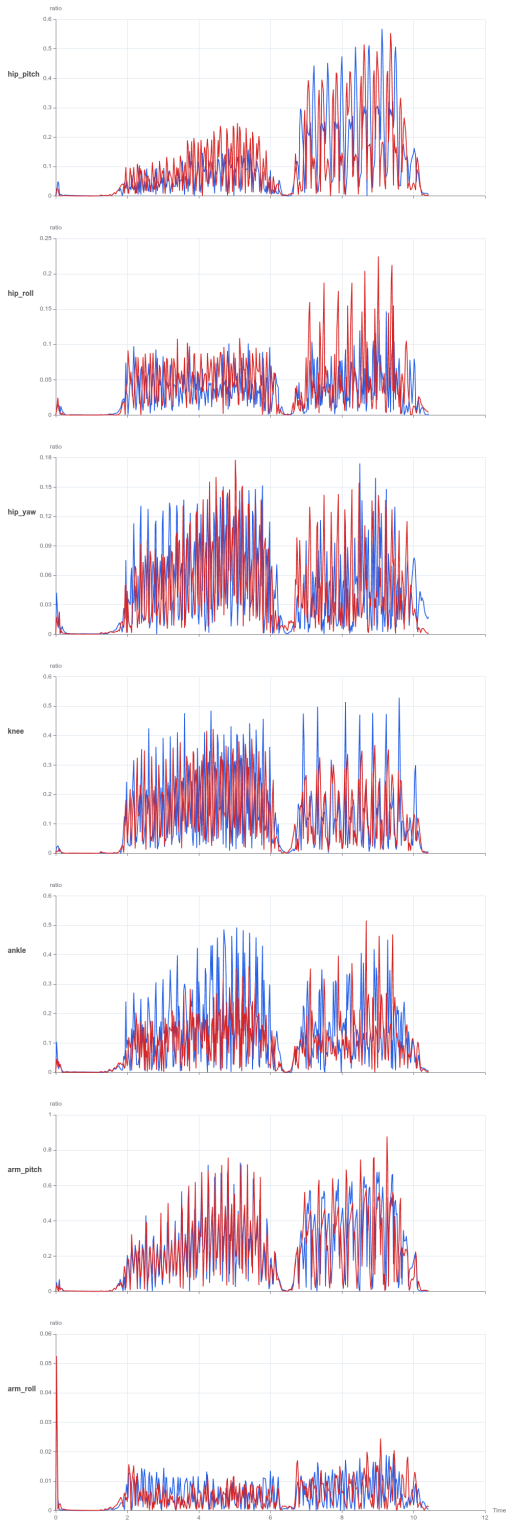

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用

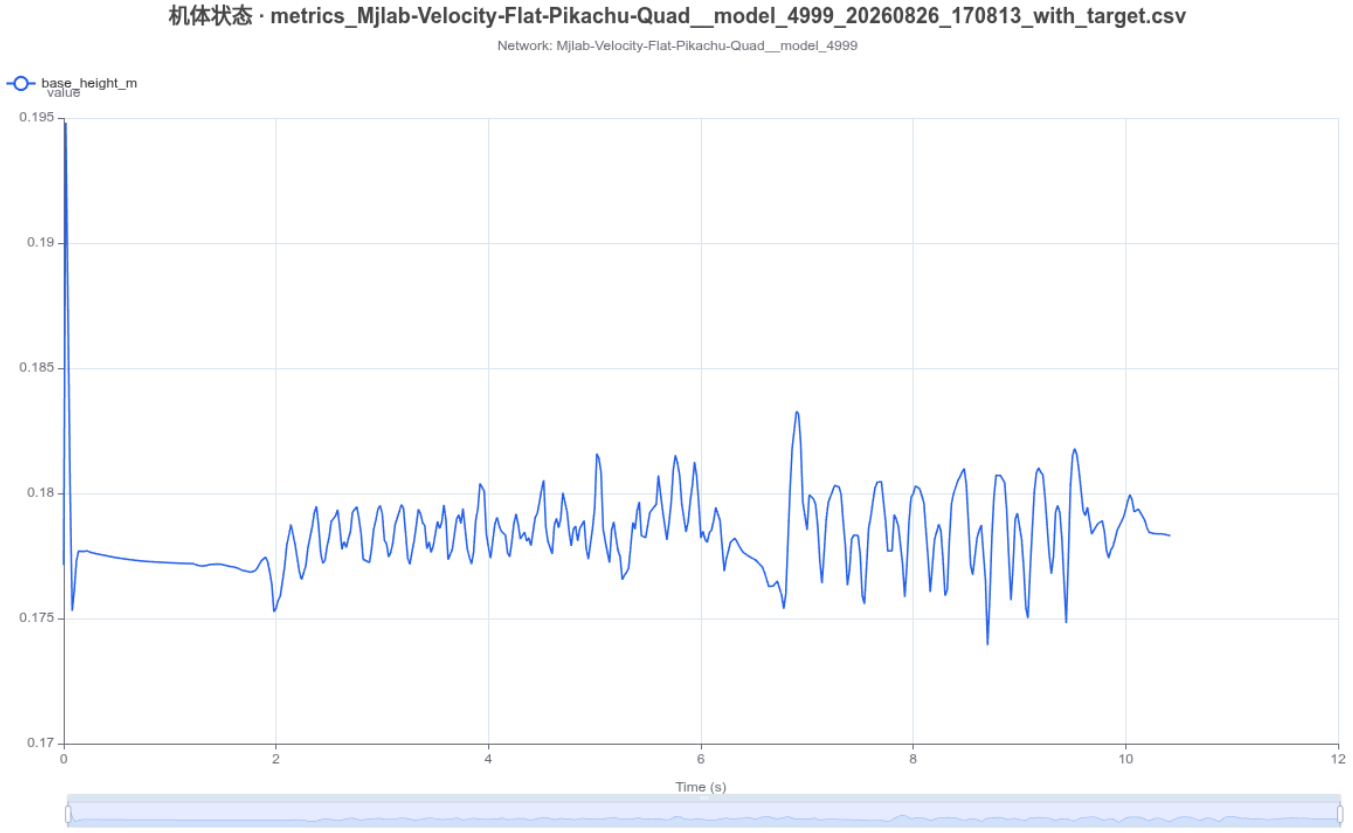


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	470	0.00000110593	0.567730	0.0957667	0.147926	0.567730	0.320019
joint_velocity_ratio/left_hip_roll_joint	10	15	470	5.81988e-9	0.146815	0.0313067	0.0420675	0.146815	0.0823521
joint_velocity_ratio/left_hip_yaw_joint	10	15	470	6.83911e-9	0.174021	0.0429052	0.0603079	0.174021	0.127777
joint_velocity_ratio/left_knee_joint	30	10	470	4.99359e-7	0.528468	0.119466	0.172570	0.528468	0.386879
joint_velocity_ratio/left_ankle_joint	10	15	470	1.41248e-7	0.492235	0.115790	0.165209	0.492235	0.349665
joint_velocity_ratio/left_arm_pitch_joint	10	15	470	0.00000214096	0.729271	0.208549	0.287128	0.729271	0.595084
joint_velocity_ratio/left_arm_roll_joint	10	15	470	7.47021e-9	0.0467158	0.00516900	0.00718577	0.0467158	0.0132250
joint_velocity_ratio/right_hip_pitch_joint	30	10	470	3.06628e-7	0.553280	0.0998576	0.152325	0.553280	0.363564
joint_velocity_ratio/right_hip_roll_joint	10	15	470	3.61133e-9	0.224639	0.0394185	0.0559396	0.224639	0.105065
joint_velocity_ratio/right_hip_yaw_joint	10	15	470	1.09847e-8	0.177612	0.0414330	0.0587016	0.177612	0.127183
joint_velocity_ratio/right_knee_joint	30	10	470	3.65033e-7	0.421996	0.118402	0.162786	0.421996	0.329814
joint_velocity_ratio/right_ankle_joint	10	15	470	7.57735e-8	0.515777	0.0929890	0.128386	0.515777	0.256441
joint_velocity_ratio/right_arm_pitch_joint	10	15	470	0.00000184923	0.877871	0.208837	0.286910	0.877871	0.570774
joint_velocity_ratio/right_arm_roll_joint	10	15	470	2.26877e-9	0.0524520	0.00474941	0.00684983	0.0524520	0.0130487

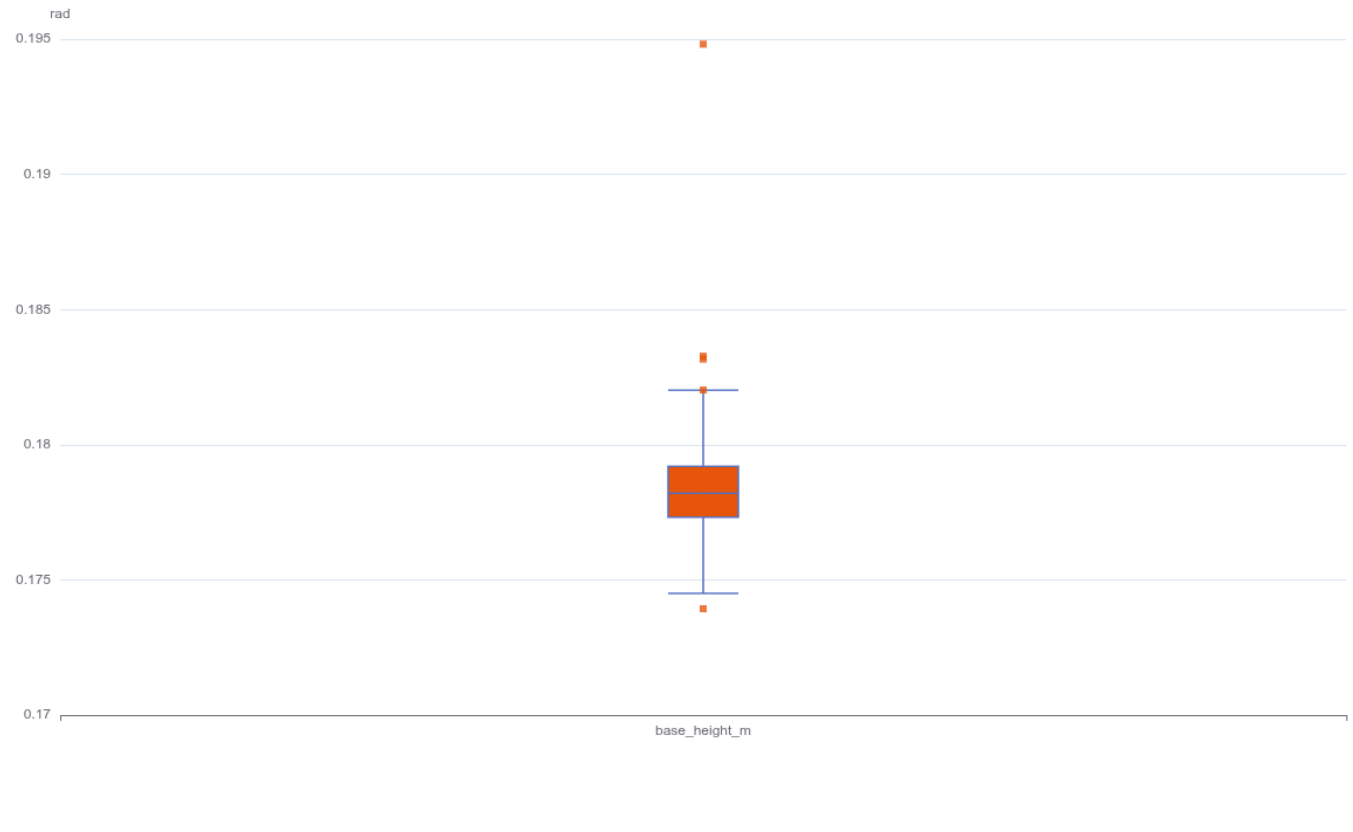

ratio
1
0.8
0.6
0.4
0.2
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

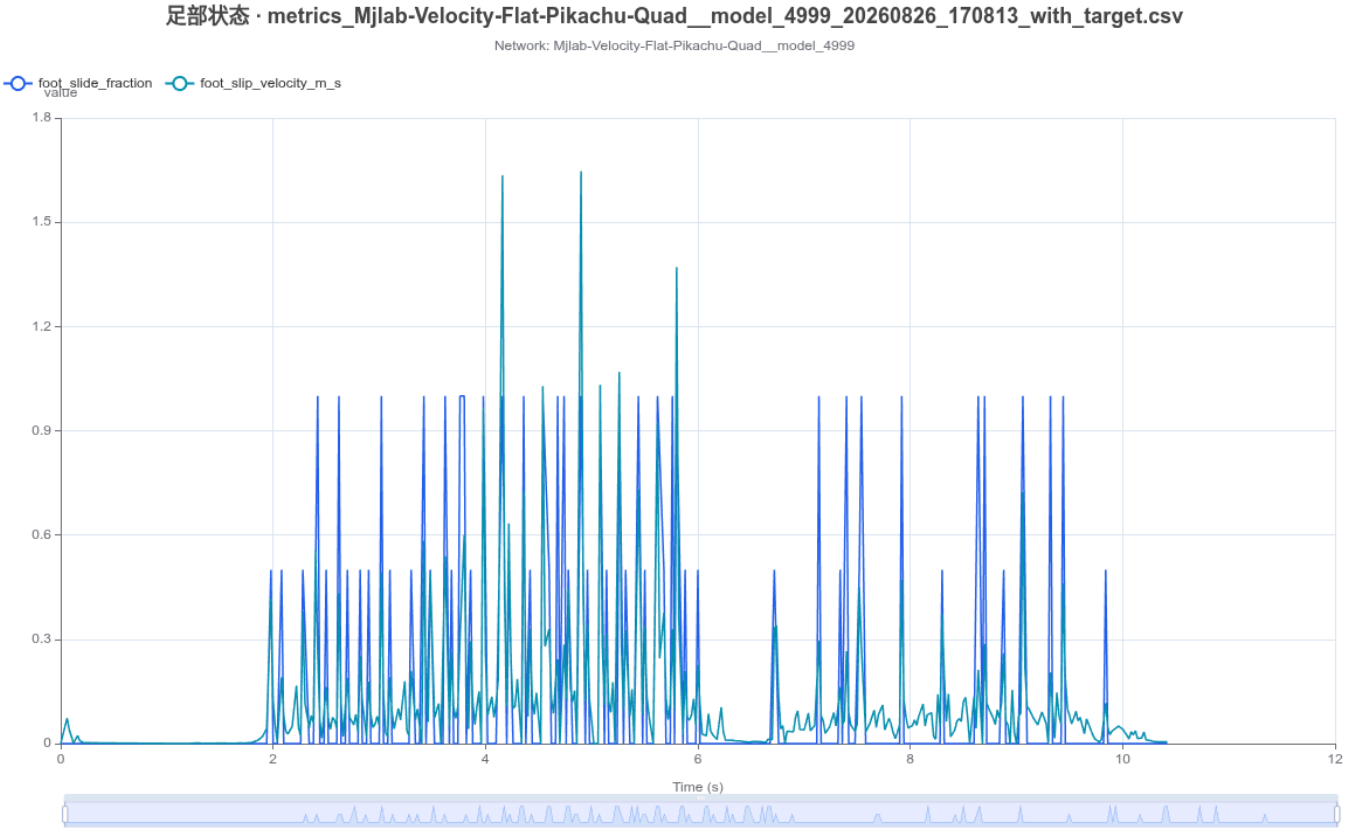
机体状态



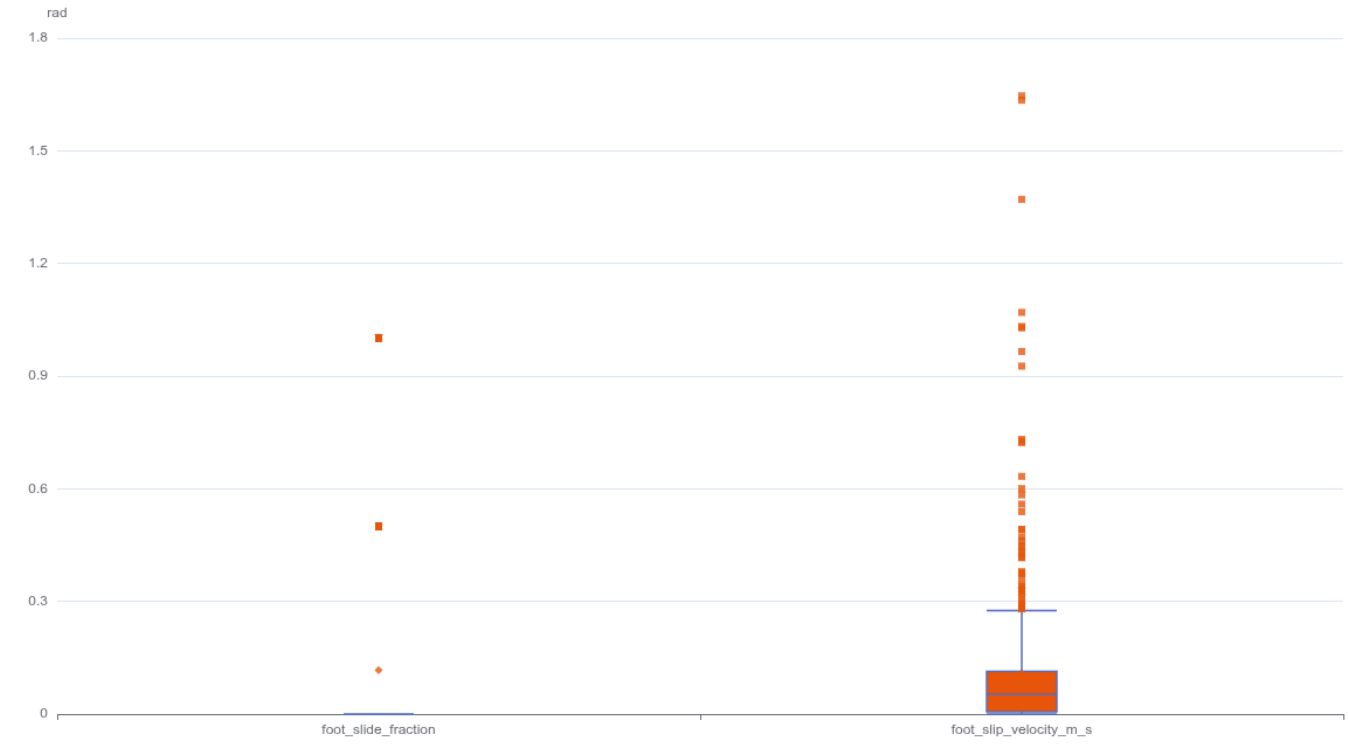
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	470	0.173933	0.194813	0.178349	0.178356	0.194813	0.180717



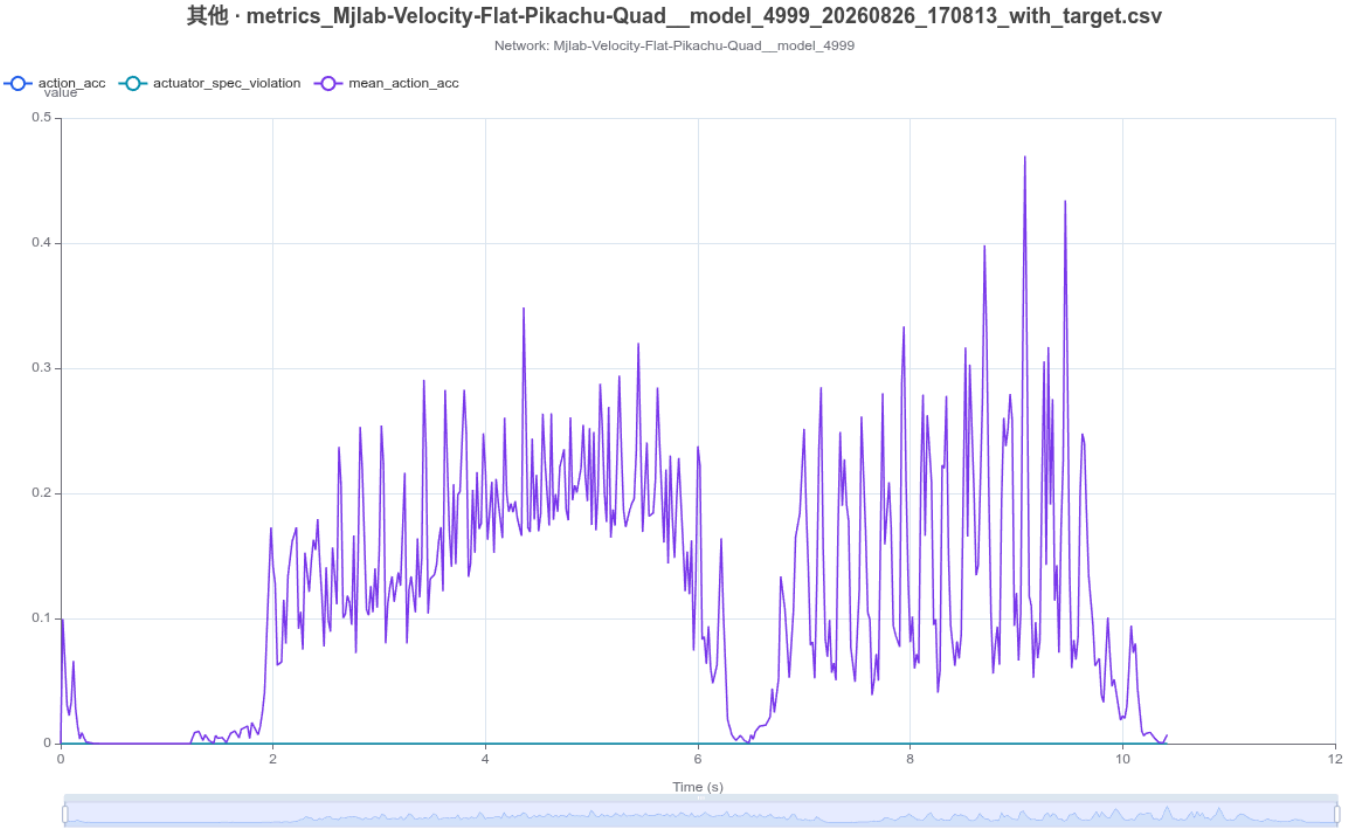
足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	470	0.00000	1.00000	0.117021	0.314541	1.00000	1.00000
foot_slip_velocity_m_s	–	–	470	0.00000	1.64629	0.108264	0.220707	1.64629	0.417496



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	470	6.22656e-8	0.469649	0.120024	0.152362	0.469649	0.278922
actuator_spec_violation	–	–	470	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
mean_action_acc	–	–	470	6.22656e-8	0.469649	0.120024	0.152362	0.469649	0.278922

